Temporal Measurement Instability in LLM Benchmarks: A Large-Scale Psychometric Audit of 5,227 Models

Anonymous ACL submission

Abstract

LLM benchmark scores are routinely compared across model generations, yet whether individual items measure the same construct over time is rarely tested—and existing contamination detectors validate against ground truth that conflates global exposure with differential functioning. We develop a semi-synthetic injection framework with cross-subject matching that achieves zero false positive inflation under single-subject contamination ($\Delta\text{FPR} = 0.000$; $\Delta\text{FPR} < 0.01$ for $k \leq 5$ subjects) while preserving detection power under worst-case uniform injection ($\Delta\text{TPR} = +0.508$ at $p = 0.5$), and show that web-overlap contamination labels are near-independent of Mantel-Haenszel Differential Item Functioning (DIF) flags (enrichment ≈ 0.96, $\phi = 0.012$)—exposing a category error in prevailing validation methodology. Applying MH-DIF with ETS severity classification to six METABENCH benchmarks (5,227 models, 27,125 items), we find 17.9–31.3% of items exhibit large temporal DIF (ETS C) between 2023 and 2024 cohorts, each 45–174× above random baselines, with knowledge/reasoning benchmarks more affected (C% $\geq 26.9\%$) than procedural ones (C% $\leq 18.9\%$). This does not threaten aggregate ranking validity ($\rho \approx 0.997$, $\tau \approx 0.958$), but specific model families show concentrated rank shifts (Llama-3: $+53$ median positions, Mistral: -35) and 2.0% of pairwise rankings reverse. Item-level decomposition shows 92.4% of the signal is item-specific; DIF-flagged items concentrate discriminative information ($z = -8$ to -49) with a difficulty–direction interaction ($\rho = -0.597$): gains on hard items, erosion on easy ones. LLM benchmarks need psychometric monitoring, not just score reporting.

1 Introduction

When we say "Model B outperforms Model A on MMLU," we implicitly assume that the benchmark measures the same construct in the same way for both models. This assumption of temporal comparability underpins every cross-generation comparison published on leaderboards and in model reports, yet it is rarely tested. We apply Mantel-Haenszel Differential Item Functioning (MH-DIF) analysis (Holland and Thayer, 1988) with Educational Testing Service (ETS) severity classification to the METABENCH response matrix (Kipnis et al., 2024),[1] and find that 31% of MMLU (Hendrycks et al., 2021) items receive ETS Category C classification ($|\Delta_{\text{MH}}| \geq 1.5$, $p < 0.05$), indicating large shifts in item functioning between models released in 2023 and 2024. This is not statistical noise: a random-split control within the 2023 cohort yields only 0.18% ETS C items. This pattern persists across all 57 MMLU subjects (C% range 21–48%, unweighted mean $\approx 30\%$), and item-level regression shows that 92.4% of the DIF signal is unrelated to aggregate domain-level accuracy gains. Nearly one in three MMLU items behaves differently depending on when the evaluated models were trained. Crucially, this measurement instability does not threaten benchmark validity: Spearman rank correlation between model rankings computed on all items versus stable-only items is $\rho \approx 0.997$. The instability is diagnostic, not destructive; it reveals which capability dimensions shift across model generations rather than rendering MMLU scores unreliable.

A growing body of contamination detection research has produced sophisticated item-level methods, from membership inference approaches such as Min-K% (Shi et al., 2024) to performance-based detectors such as ConStat (Dekoninck et al., 2024), DVD (Liang, 2026), and Self-Critique (Tao et al., 2026). These methods validate detection accuracy primarily against web-overlap labels such as those of Li et al. (2024b), which classify bench-

[1] Analysis code and DIF audit scripts are available at `https://anonymous.4open.science/r/dif-audit`.

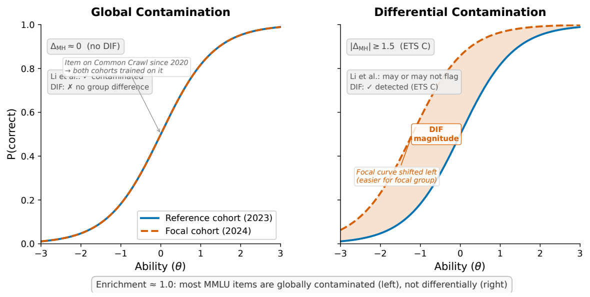

Figure 1: Global vs. differential contamination, illustrated via Item Characteristic Curves (ICCs). **Left**: global contamination produces no DIF; both cohorts encounter the item, and their ICCs overlap. **Right**: differential contamination produces DIF; the focal cohort's ICC shifts because only that cohort was disproportionately exposed during training. The enrichment ratio of ≈ 1.0 between DIF flags and web-overlap labels is expected because most MMLU items fall into the left category.

mark items as contaminated based on verbatim or near-verbatim matches in publicly accessible training corpora. Such labels measure *global exposure*: whether an item exists in the data ecosystem that all training pipelines can access. They do not measure *differential item functioning*: whether an item behaves differently across model cohorts that may have had varying degrees of exposure or different training methodologies. Figure 1 illustrates the distinction. Most MMLU items are globally contaminated in the sense that they exist in web-accessible text, so all model cohorts encounter them and no differential signal arises. The items that function differentially are those where exposure or training methodology changed asymmetrically across cohorts. Validating differential detectors against global-exposure labels therefore conflates two phenomena. Direct comparison confirms near-independence between the two flag sets (enrichment $= 0.964$, Jaccard $= 0.206$, $\phi = 0.012$), consistent across contamination subtypes.

We apply classical MH-DIF with ETS classification to the full METABENCH response matrix (5,227 models $\times$ 12,508 MMLU items), to our knowledge, the largest-scale DIF analysis on LLM benchmarks. This psychometric approach complements concurrent work on anchor-item calibration for score comparability (Habba et al., 2026) and addresses the fragility of existing contamination detectors under post-training perturbation (Wang et al., 2026). Semi-synthetic injection experiments confirm that MH-DIF is sensitive to differential item behavior when present (ΔTPR $= +0.508$ at exploitation rate $p = 0.5$), and our cross-subject matching protocol eliminates false positive inflation (ΔFPR $= 0.000$). Three structural controls (matched-ability quintile analysis, IRT θ-conditioning, and within-subject pre-checks) confirm that the approximately 31% temporal DIF rate cannot be attributed to methodological artifacts, establishing it as a robust empirical finding about LLM benchmark measurement properties, replicated across all six METABENCH benchmarks (Section 4.7). MH-DIF detects *that* items function differently across cohorts, not *why*—the signal may reflect contamination, capability change, or training methodology shifts.

MMLU's 12,508 items provide sufficient measurement redundancy that removing nearly a third does not alter rank order ($\rho \approx 0.997$). DIF should therefore be understood as a diagnostic lens rather than a quality filter: DIF-flagged items concentrate discriminative signal ($z = -8$ to -49 vs. random removal), and model families show distinctive DIF profiles (Llama-3 gains $+53$ median rank positions, Mistral loses 35).

Our contributions are methodological and empirical, not algorithmic:

1. **Ground truth meta-critique**: Web-overlap contamination labels (Li et al., 2024b) do not provide appropriate ground truth for item-level differential detectors. Direct empirical comparison on 9,229 items shows near-independence (enrichment $= 0.964$, Jaccard $= 0.206$, $\phi = 0.012$), exposing a validation methodology gap (Section 4.3).

2. **Semi-synthetic validation framework**: Controlled injection with cross-subject matching achieves ΔFPR $= 0.000$ while preserving detection power (ΔTPR $= +0.508$ at $p = 0.5$); cross-benchmark replication on GSM8K confirms generality (Sections 4.5 and 4.7).

3. **Temporal measurement instability at scale**: Approximately 31% of MMLU items exhibit differential functioning across model generations under ETS C classification; replication across all six METABENCH benchmarks yields C% $= 17.9$–31.3% (45–174$\times$ above random baselines), with knowledge/reasoning benchmarks showing higher DIF rates than procedural ones. DIF-flagged items concentrate discriminative signal ($z = -8$ to -49

vs. random removal) and exhibit a difficulty–direction interaction ($\rho = -0.597$): gains on hard items, erosion on easy ones (Sections 4.2, 4.6 and 4.7).

2 Related Work

Contamination Detection. A growing family of methods detects benchmark contamination through model behavior or training data analysis. Membership inference approaches estimate whether a model has seen specific items during training: Min-K% (Shi et al., 2024) and Min-K%++ (Zhang et al., 2024) use token-level log-likelihood statistics, while Time Travel (Golchin and Surdeanu, 2024) probes models with temporal cues and Oren et al. (2024) prove contamination through exchangeability tests. Performance-based detectors identify contamination through statistical anomalies (Dekoninck et al., 2024; Liang, 2026; Tao et al., 2026; Deng et al., 2024). Balloccu et al. (2024) provide a broader taxonomy of leakage detection strategies. These methods share a common validation practice: they verify detection accuracy against web-overlap or n-gram labels (Li et al., 2024b) that classify items based on verbatim matches in publicly accessible corpora. Such labels measure *global exposure* (whether an item exists in training data) rather than *differential item functioning* (whether an item behaves differently across model cohorts). This global-vs-differential distinction motivates our ground truth analysis (Section 4.3). Work on LLM memorization (Carlini et al., 2023; Tirumala et al., 2022) establishes that memorization depends on data frequency and model scale, reinforcing the distinction between global exposure and differential behavior. Wang et al. (2026) show that existing contamination detectors are fragile under reinforcement learning, further motivating behavioral approaches grounded in psychometric theory.

IRT in LLM Evaluation. IRT adoption in LLM evaluation has accelerated: Lalor et al. (2019) introduced IRT to NLP, Vania et al. (2021) assessed test set discriminability, and Rodriguez et al. (2021) demonstrated that not all evaluation examples are equally informative. Recent extensions include PSN-IRT (Zhou et al., 2026) for benchmark quality at scale, TinyBenchmarks (Maia Polo et al., 2024) for IRT-based item selection, scalability coefficients for detecting bad benchmark items (Hardy et al., 2026), ATLAS (Li et al., 2025) for computerized adaptive testing of LLMs, and competency-focused IRT evaluation in medical domains (Luo et al., 2025). The closest concurrent work is Growing Pains (Habba et al., 2026), which addresses score comparability through fixed-parameter calibration (FPC; Kim and Kolen, 2019). Our work is complementary: FPC methods assume a subset of anchor items with stable measurement properties; temporal DIF analysis directly tests this stability assumption and can identify candidate anchor items through DIF pre-screening, providing input to equating procedures (Kolen and Brennan, 2014). FPC assumes a stable set of anchor items; our finding that 31% of items exhibit temporal DIF directly informs anchor selection by identifying which items should *not* serve as anchors. Empirical comparison awaits public release of anchor item lists.

DIF Methodology and Measurement Invariance. The Mantel-Haenszel procedure for DIF detection was introduced by Holland and Thayer (1988) and remains the most widely used method in educational testing due to its simplicity and well-characterized statistical properties (Clauser and Mazor, 1998). Alternative approaches include logistic regression DIF (Swaminathan and Rogers, 1990), SIBTEST (Shealy and Stout, 1993), Lord's χ^2 test (Lord, 1980), and the Raju area method (Raju, 1988); Zumbo (1999) provides a comprehensive comparison. DIF analysis is a specific instance of measurement invariance testing (Meredith, 1993; Vandenberg and Lance, 2000), which examines whether a measurement instrument functions equivalently across populations. MIMIC models offer a latent-variable approach to the same problem (Muthén, 1985; Woods, 2009), and Teresi and Fleishman (2007) reviews DIF methods across applied domains. Cross-subject matching—computing the stratification variable from domains other than the one being tested—has psychometric precedent in the DIF literature (Nandakumar, 1993; Donoghue and Allen, 1993); our adaptation addresses LLM-specific challenges of multi-domain local dependence and simultaneous contamination control at scale. We do not modify the MH procedure; our contribution is the application context and the empirical findings it reveals. The scale of this analysis (5,227 examinees across a 57-subject multidimensional ability space) exceeds typical educational testing applications, where Belzak (2020) note that DIF methods can behave differently at extreme sample sizes. The primary methodological challenge is the impact-versus-DIF confound: genuine ability differences between temporal cohorts can inflate DIF estimates. We address this through matched-ability quintile analysis, IRT θ-conditioning, and a random-split control (Section 4.2).

Benchmark Quality and Design. The Benchmark Health Index (Zhu et al., 2026) and Benchmark[2] (Qian and Huang, 2026) assess static benchmark quality; our temporal DIF analysis is complementary, diagnosing *when* measurement properties shift rather than evaluating them at a single snapshot. Maeda (2025) predict which item features associate with DIF in educational settings; we instead detect DIF empirically across LLM cohorts at scale. Dynamic benchmarks (Kiela et al., 2021; Li et al., 2024a; White, 2025) mitigate contamination by construction, yet none provides psychometric equating across versions. Chatbot Arena (Chiang et al., 2024) sidesteps static benchmarks through pairwise preference. Broader surveys (Chang et al., 2024; Burnell et al., 2023) identify benchmark saturation and lack of item-level transparency as systemic challenges that our DIF analysis directly addresses.

3 Method

We adopt Differential Item Functioning (DIF) analysis from educational measurement to assess whether individual benchmark items behave consistently across groups of LLMs. The procedure requires only a binary response matrix and group labels, with no access to model internals or training data. Below we define the data, the statistical test, the cohort designs, and a semi-synthetic validation framework.

3.1 Data

The primary dataset is the METABENCH response matrix (Kipnis et al., 2024), which records binary outcomes (1 = correct, 0 = incorrect) for 5,227 models evaluated on 12,508 MMLU items (Hendrycks et al., 2021). Models span release dates from 2023 to 2024, drawn from the Open LLM Leaderboard. MMLU covers 57 subjects grouped into five broad domains (STEM, Humanities, Social Sciences, Other, and a mixed category). We supplement the response matrix with the web-overlap contamination labels of Li et al. (2024b), which classify 9,229 of 12,508 items (73.8% coverage) as "contaminated" or "clean" based on verbatim matches in publicly accessible web corpora. These labels serve as an external reference for evaluating the relationship between DIF flags and existing contamination indicators.

3.2 Mantel-Haenszel DIF and ETS Classification

In educational testing, Differential Item Functioning (DIF) occurs when an item's statistical relationship to the latent trait θ differs between two groups after conditioning on ability level. An item with DIF is not merely harder for one group; it functions differently relative to what the group's overall ability would predict. The standard terminology refers to test-takers as "examinees" and questions as "items." In our setting, examinees are LLMs and items are benchmark questions.

The Mantel-Haenszel (MH) procedure (Holland and Thayer, 1988) detects DIF by stratifying examinees into K score bins based on total score (a proxy for θ) and computing a common odds ratio across strata. We use $K = 5$ equiprobable strata (quintile-based binning) following ETS recommended practice for large-scale assessments (Donoghue and Allen, 1993; Clauser and Mazor, 1998). Within each stratum k, the responses of the reference group and the focal group form a 2×2 contingency table:

$$\alpha_{\mathrm{MH}} = \frac{\sum_{k=1}^{K} A_k D_k \,/\, T_k}{\sum_{k=1}^{K} B_k C_k \,/\, T_k}, \qquad (1)$$

where A_k and B_k are the number of correct and incorrect responses in the reference group for stratum k, C_k and D_k are the corresponding counts for the focal group, and T_k is the total number of examinees in stratum k. Under the null hypothesis of no DIF, $\alpha_{\mathrm{MH}} = 1$. The log odds ratio is converted to the ETS delta scale:

$$\Delta_{\mathrm{MH}} = -2.35 \ln(\alpha_{\mathrm{MH}}) \,. \qquad (2)$$

The Educational Testing Service (ETS) classifies DIF severity into three categories based on $|\Delta_{\mathrm{MH}}|$. Category A (negligible) requires $|\Delta_{\mathrm{MH}}| < 1.0$. Category C (large) requires both $|\Delta_{\mathrm{MH}}| \geq 1.5$ and statistical significance at $p < 0.05$ from the MH chi-squared test. Category B (moderate) includes all remaining items: those with $1.0 \leq |\Delta_{\mathrm{MH}}| <$

1.5, or $|\Delta_{\mathrm{MH}}| \geq 1.5$ without statistical significance. The dual threshold in ETS C (effect size *and* significance) guards against flagging items that show large but unreliable effect sizes. Throughout this paper, we report the percentage of items classified as ETS C, denoted C%. For example, if 2024 models consistently outperform 2023 models on a formal-logic item *after controlling for overall ability*, the pooled odds ratio $\alpha_{\mathrm{MH}} = 0.42$ yields $\Delta_{\mathrm{MH}} = -2.35 \ln(0.42) = 2.04$, classified as ETS C ($|\Delta_{\mathrm{MH}}| \geq 1.5$, $p < 0.001$).

The choice of matching variable (the total score used for stratification) affects both validity and interpretation. We distinguish three matching strategies: (1) *standard matching* uses the within-benchmark total score on all items of the same benchmark; (2) *cross-subject matching*, adapting purified subtest matching (Swaminathan and Rogers, 1990) to the LLM multi-subject setting, uses the total score on all subjects *other than* the one containing the item under test, preventing the matching variable from absorbing injected contamination signals (Section 3.4); and (3) *external matching* uses the total score on a different benchmark entirely (e.g., MMLU total score as the matching variable when testing DIF on GSM8K items), providing an independent ability proxy uncontaminated by the target benchmark. Cross-subject and external matching both address matching variable contamination but at different granularities: cross-subject operates within a multi-domain benchmark, while external operates across benchmarks. Total score matching is sufficient: IRT θ correlates at $r = 0.9986$ with total scores (Section 4.4).

LLM-specific challenges. Applying MH-DIF to LLMs introduces assumptions absent in educational testing. First, models trained on overlapping corpora violate local independence; permutation simulation (1,000 iterations $\times$ 8 subjects, Q3 $= 4.6$–34.2%) confirms LD-attributable FPR inflation < 0.001 pp under cross-subject matching with ETS C (Section A). Second, fine-tuned variants and merges within model families are not independent examinees; graduated de-duplication shows family structure does not inflate C% (Section 4.4). Third, the ability distribution is non-normal (bimodal from base vs. instruction-tuned models), motivating quintile-based rather than parametric stratification. Fourth, MMLU's 57 subjects span multiple latent dimensions, while MH assumes unidimensionality; cross-subject matching and within-subject DIF checks address this (Section 4.4). MH-DIF is non-parametric and does not assume a specific IRT model.

3.3 Cohort Designs

We construct five cohort comparisons: (1) *Temporal*: 2023 ($N = 1{,}080$) vs. 2024 ($N = 807$) models, the primary comparison; (2) *Ability-quintile*: Q1–Q2 vs. Q4–Q5 within the same temporal window, testing whether ability differences alone produce DIF; (3) *Within-family*: Llama-2 ($N = 161$) vs. Llama-3 ($N = 308$), isolating generational changes within a single family; (4) *Within-subject*: per-subject DIF to test domain concentration; and (5) *Random split*: two random halves of the 2023 cohort ($N = 540$ each) as a sanity check.

3.4 Semi-Synthetic Injection Framework

To evaluate whether MH-DIF can detect differential contamination when it is present, we construct a semi-synthetic framework with controlled ground truth.

Procedure. From items classified as ETS A in the uninjected data (negligible DIF), we select 20% uniformly at random as the "contaminated" set. For each contaminated item, we flip incorrect responses to correct in the focal group at exploitation rate $p \in \{0.0, 0.3, 0.5, 0.7, 1.0\}$. We then run MH-DIF on the modified response matrix and measure detection performance. We report $\Delta\mathrm{TPR} = \mathrm{TPR}(\text{injected}) - \mathrm{TPR}(\text{baseline})$ and $\Delta\mathrm{FPR} = \mathrm{FPR}(\text{injected}) - \mathrm{FPR}(\text{baseline})$, where TPR is the true positive rate on injected items and FPR is the false positive rate on uninjected items. Each injection level is repeated across 5 random seeds for uncertainty estimation. Because binary response flipping represents a worst-case uniform exploitation pattern, reported $\Delta\mathrm{TPR}$ values constitute upper-bound estimates of detection sensitivity under more realistic contamination scenarios.

Cross-subject matching. The choice of matching variable is critical for valid semi-synthetic evaluation. When contamination is injected into items within subject X, the total score on subject X absorbs part of the injection signal. Using this contaminated total score as the MH matching variable violates the conditional independence assumption and inflates FPR. Cross-subject matching resolves this by computing the matching variable from all subjects *other than* the one being tested.

Remark (Zero FPR under single-subject matching). Let $M_i^{(-s)} = \sum_{j \notin s} Y_{ij}$ be the matching variable for examinee i on focal subject s. Under single-subject injection (contamination confined to subject s), the tested item belongs to s and does not contribute to $M_i^{(-s)}$; injecting contamination into s leaves $M_i^{(-s)}$ unchanged. Therefore $\Delta\text{FPR} = 0$ by construction.

When contamination spans multiple subjects simultaneously, $M_i^{(-s)}$ may absorb signal from other contaminated subjects, potentially inflating FPR. Empirically, this guarantee extends to multi-domain contamination affecting ≤9% of subjects ($k \leq 5$), with graceful degradation beyond this boundary (Section 4.5).

Empirically, $\Delta\text{FPR} = 0.000$ across all five MMLU domains (bootstrap 95% CI $= [0.000, 0.000]$). $\Delta\text{FPR} = 0$ is a constructive guarantee of the matching strategy, independent of injection realism, whereas ΔTPR constitutes an upper-bound estimate. We adopt cross-subject matching as the standard protocol.

Diagnostic procedures (Yen's Q3 local independence, domain-specificity analysis, DIF decomposition) and statistical details (multiple testing correction, threshold sensitivity, architecture heterogeneity checks) appear in Section A.

4 Experiments

We apply MH-DIF to the METABENCH response matrix (Kipnis et al., 2024) (5,227 models × 12,508 MMLU items (Hendrycks et al., 2021)), with temporal cohorts defined by release date (2023 vs. 2024). Cross-benchmark validation across all six METABENCH benchmarks appears in Section 4.7.

4.1 Validation: DIF Detection Power

As approximate parametric bounds, a Monte Carlo power analysis (1,000 replications per condition) using empirical 2PL parameters from PSN-IRT (Zhou et al., 2026) ($\bar{a} = 1.2$, $\bar{b} = -0.84$; 14,042 MMLU items) confirms that at $N \geq 80$ per cohort, ETS C achieves TPR of 0.68–0.80, FPR of 0.036–0.042, and AUC of 0.93–0.94 for items with $|\Delta_{\text{MH}}| \geq 1.4$ (details in Section A). Our temporal cohorts ($N_{\text{ref}} = 1{,}080$, $N_{\text{focal}} = 807$) far exceed this threshold. Power varies substantially with item discrimination: TPR ranges from 0.40 at $a = 0.5$ to 0.95 at $a = 2.0$; the gap between TPR at mean a (0.87) and the aggregate TPR (0.68) quantifies the power cost of parameter heterogeneity, consistent with Jensen's inequality (Section A.2).

4.2 Temporal Comparability: 31% Measurement Instability

We compare 2023 ($N = 1{,}080$) against 2024 ($N = 807$) models; 31.3% of items receive ETS C classification ($|\Delta_{\text{MH}}| \geq 1.5$; 95% bootstrap CI: [29.4%, 38.1%], $B = 1{,}000$ model-level resamples; rightward asymmetry reflects heavy-tailed C% from subjects exceeding 40%). Benjamini-Hochberg FDR correction at $q = 0.05$ removes only 3 items (31.32% → 31.30%), because the dual ETS threshold already provides implicit multiple-testing protection.

Three controls confirm this reflects genuine temporal change. A random split of the 2023 cohort yields only 0.18% ETS C items (sampling noise), and an ability-quintile comparison within the same temporal cohort yields 2.9% (ability differences alone). A within-family comparison (Llama-2 vs. Llama-3) produces 61.3% ETS C items, elevated by large ability gaps (Holland and Thayer, 1988). The temporal cohort ability gap is negligible: Cohen's $d = -0.17$ with 76.4% distributional overlap. Under 1:1 nearest-neighbor matching on total score ($N = 638$ per group, 5 seeds), C% $= 28.9\% \pm 0.1\%$, confirming at most ∼2.4 pp is attributable to ability mismatch. Permutation tests confirm ($p < 0.01$; Section A).

Across all 57 subjects, C% ranges from 20.7% to 48.4% (mean ≈ 30%; Table 1). DIF rates are domain-invariant (Kruskal-Wallis $p = 0.78$), but DIF *direction* exhibits domain dependence: STEM items favor 2024 models (55.4%), Humanities favor 2023 (64.4%). Item-level regression shows 92.4% of the DIF signal is item-specific (pseudo $R^2 = 0.0007$). A Simpson's paradox confirms that DIF is associated with item properties rather than aggregate accuracy gains: domain accuracy positively predicts DIF at the aggregate level (OR $= 1.55$, $p = 0.001$) but reverses after controlling for item difficulty and discrimination (OR $= 0.73$, $p = 0.030$). A difficulty–direction interaction emerges ($\rho = -0.597$): 2024 models gain on hard items while showing relative erosion on easy ones. The conclusion is robust across thresholds (18.3% at $|\Delta_{\text{MH}}| \geq 2.0$, 10.2% at ≥ 2.5).

4.3 Ground Truth Mismatch

Direct comparison of DIF flags against the web-overlap labels of Li et al. (2024b) on 9,229

Table 1: DIF landscape across cohort definitions. ETS C denotes the percentage of items with $|\Delta_{\text{MH}}| \geq 1.5$. FPR is the flagging rate on web-overlap "clean" items (Li et al., 2024b), not a traditional false positive rate. Enrichment is the ratio of observed to expected overlap between DIF flags and web-overlap labels. Dashes indicate comparisons where Li et al. labels are not applicable.

Cohort Definition	N_{ref} / N_{focal}	C%	FPR (clean)	Enrich.
Temporal (2023 vs. 2024)	1,080 / 807	31.3	0.318	0.964
Ability quintile (Q1–2 vs. Q4–5)	∼2,600 / ∼2,600	2.9	0.027	1.19
Within-family (Llama-2 vs. 3)	161 / 308	61.3	–	–
Random split (sanity)	540 / 540	0.18	–	–
Within-subject (avg, 5 subj.)	*varies*	38.1	–	–

annotated items confirms that the two methods capture orthogonal signals. The enrichment ratio is 0.964 (Table 1): DIF-C rates are 30.7% among web-contaminated items and 31.8% among clean items. Association tests yield $\phi = 0.012$, $\chi^2 = 1.28$ ($p = 0.258$), and OR $= 1.05$ [0.96, 1.15]; the Jaccard index between flagged sets is 0.206. This near-independence holds across contamination subtypes (input-only enrichment $= 0.986$, input-and-label $= 0.960$). The null result does not indicate detection failure: semi-synthetic injection produces a clear dose-response (ΔTPR $= +0.508$ at $p = 0.5$; Figure 2). The disconnect arises because web-overlap labels measure *global exposure* (whether an item exists in training corpora), while DIF measures *differential functioning* (whether an item's relationship to ability changes across cohorts). The Li et al. labels are the wrong ground truth for a differential method.

4.4 Structural Analysis

Three structural controls fail to reduce temporal C% below 31% (Table 2): matched-ability quintiles, IRT θ-conditioning ($r(\theta, \text{total score}) = 0.9986$), and within-subject DIF (C% $= 38.1\%$; range 31.1–45.2% across 5 representative subjects). The elevated within-subject rate partly reflects local dependence: 11.3% of within-subject item pairs exceed $|Q3| > 0.20$ vs. 2.2% cross-subject, confirming that cross-subject C% $= 31.3\%$ is the more conservative estimate. The near-perfect θ–total score correlation confirms that the analysis is essentially CTT-based; IRT provides no incremental information for ability stratification in this application. Logistic regression DIF confirms 92.1% concordance with MH and reveals additional non-uniform DIF (24.8%), making 31% conservative. Graduated deduplication (K=10: 22.7%; K=20: 29.1%) confirms family structure does not inflate C%. Ability-matched analysis reduces Llama-2 vs. Llama-3 C% from 61.3% to 28.3% and temporal C% from 31.3% to 28.9%—consistent matched values (∼28–29%) despite vastly different unmatched rates, suggesting genuine item-level DIF accounts for ∼28–29% of items. Six alternative explanations were tested; none reduces C% below ∼28% (Section B).

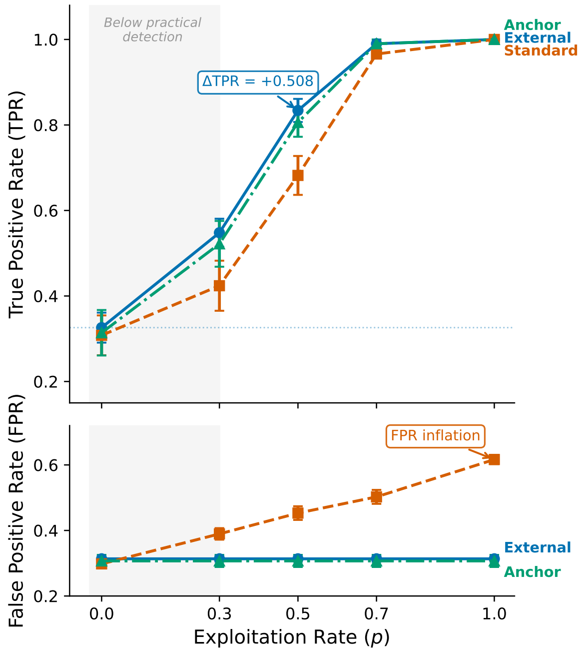

Figure 2: Semi-synthetic dose-response for DIF detection on MMLU. ΔTPR increases monotonically with exploitation rate p, confirming that MH-DIF is sensitive to differential item behavior when present. Baseline (no injection) produces $\Delta\text{TPR} \approx 0$.

Table 2: Structural approaches to reducing temporal DIF. None reduces temporal C% below 31%, indicating that the observed instability is not an artifact of ability matching.

Approach	Mechanism	Temp. C%	TPR (p=0.5)	Outcome
Baseline (ext.)	Cross-bench. total score	$0.314_{\pm.009}$	$0.834_{\pm.027}$	Reference
Matched-ability	Within-quintile temp. DIF	∼0.31	–	Enrich. 0.94–1.04
θ-cond.	θ replaces total score	0.312	0.876	$r(\theta, \text{total})$=0.9986
Within-subject	Per-subject DIF	–	–	C% $= 38.1\%$ (>25%)

4.5 Semi-Synthetic Framework Validation

Cross-subject matching achieves $\Delta\text{FPR} = 0.000$ across all five MMLU subjects (4/5 with $\Delta\text{TPR} > 0.3$ at $p = 0.5$, upper-bound estimate under uniform binary flipping; Section 3.4), while within-subject matching inflates FPR by $+0.186$ on average. Multi-subject injection ($k \in \{1, 3, 5, 10, 20\}$ subjects, 5 random draws each) confirms robustness: $\Delta\text{FPR} < 0.007$ for $k \leq 5$ ($\leq$9% of domains); at $k = 10$ (18%), ΔFPR rises to 0.026–0.069, marking the boundary of matching variable integrity (Section A and fig. 3).

4.6 Downstream Impact

Stable-item rank preservation holds across all six benchmarks (mean $\rho = 0.997$, $\tau = 0.958$; pairwise flip rate $\leq 2.8\%$; Table 3): aggregate ordering is preserved despite substantial measurement instability. DIF-flagged items concentrate discriminative signal ($z = -8$ to -49 vs. random removal; circularity ablation in Figure 4). Model families show distinctive DIF profiles: Llama-3 derivatives ($n = 315$) gain $+53$ median rank positions, while Mistral derivatives ($n = 619$) lose 35 (Figure 4). Among the 200 most displaced models, 33.6% of pairwise rankings reverse—for instance, Qwen1.5-14B-Chat ranks 610 positions ahead of mistral-7b-dpo-v6 on all items but 501 positions behind on stable items alone. Removing C-flagged items disproportionately disadvantages 2024 models (cohort gap $= 96.2$ positions, $40\times$ larger than random removal; $p < 0.001$), confirming DIF items carry concentrated inter-generational signal. That $\rho \approx 0.997$ coexists with these local reversals illustrates a general pattern: aggregate stability can mask substantial model-level instability. MMLU's 12,508 items provide sufficient redundancy; even the smallest benchmark (TruthfulQA, 817 items, 574 stable) preserves $\rho = 0.997$.

4.7 Cross-Benchmark Validation

Temporal DIF extends across all six METABENCH benchmarks (Table 3). C% ranges from 17.9% (HellaSwag (Zellers et al., 2019)) to 31.3% (MMLU), each 45–174× above random baselines. Knowledge/reasoning benchmarks (WinoGrande (Sakaguchi et al., 2020), TruthfulQA (Lin et al., 2022), ARC (Clark et al., 2018)) show C% = 26.9–31.3% versus 17.9–18.9% for procedural ones, consistent with knowledge-intensive items being more sensitive to training data composition shifts.

On GSM8K, semi-synthetic injection confirms detection power ($\Delta\text{TPR} = +0.514$ at $p = 0.5$; Section C); an exploratory GSM1K analysis suggests DIF captures capability change distinct from memorization ($r = -0.478$, 95% CI $[-0.71, -0.16]$, $p = 0.005$; Figure 5). DIF remains substantial even within a 6-month window (2023-H2 vs. 2024-H1: MMLU C% $= 34.35\%$, TruthfulQA $= 31.33\%$).

5 Discussion

5.1 Implications for Benchmark Temporal Comparability

Temporal DIF decomposes benchmark differences into three layers: (1) *impact*—ability

Table 3: Temporal DIF across six METABENCH benchmarks. C% = ETS C items ($|\Delta_{\text{MH}}| \geq 1.5$) between 2023 and 2024 cohorts; ρ/τ = Spearman/Kendall correlation between full and stable-item rankings; Flip = pairwise rank-flip rate. All six show substantial DIF (45–174× above random baseline), with knowledge/reasoning benchmarks exhibiting higher C% than procedural ones.

Benchmark	Items	C%	ρ	τ	Flip%	Rand. C%	Ratio
MMLU	12,508	31.3	0.9965	0.9600	2.00	0.18	174×
WinoGrande	1,267	30.3	0.9943	0.9432	2.84	0.32	95×
TruthfulQA	817	29.7	0.9971	0.9586	2.07	0.37	80×
ARC	1,172	26.9	0.9962	0.9538	2.31	0.60	45×
GSM8K	1,319	18.9	0.9993	0.9843	0.78	0.30	63×
HellaSwag	10,042	17.9	0.9986	0.9722	1.39	0.28	64×

(2) *DIF*—item-level differences after ability control (∼28–29% floor, 92.4% item-specific); (3) *attribution*—contamination, capability evolution, or methodology shifts, underdetermined without training data. Regardless of attribution, temporal DIF means the same score carries different meaning across model generations.

5.2 Ground Truth Meta-Critique

The near-independence between DIF flags and web-overlap labels (Section 4.3) reflects a category error: web-overlap measures global exposure, while DIF measures differential functioning, making overlap-based labels an inappropriate ground truth for a differential method. The semi-synthetic dose-response confirms detection works when differential behavior exists; the disconnect is in the ground truth, not the method.

5.3 Recommended DIF Audit Protocol

We propose running temporal MH-DIF whenever ≥ 500 new models accumulate ($N \geq 80$ suffices per Section 4.1; 500 ensures coverage of diverse model families), with graduated action levels: (1) *Monitor* (C% $< 15\%$); (2) *Flag* (15–25%): report stable-item rankings alongside full rankings; (3) *Act* (C% $\geq 25\%$): investigate flagged items and report stable-item rankings, per MMLU-Pro (Wang et al., 2024) and MMLU-Redux (Gema et al., 2024). These thresholds adapt ETS practice (Clauser and Mazor, 1998) and are portable across populations. DIF pre-screening identifies stable anchor items for equating (Kolen and Brennan, 2014; Habba et al., 2026); temporal C% complements composite quality indices (Zhu et al., 2026).

6 Conclusion

Cross-subject matching (ΔFPR $= 0.000$) resolves the validation gap: web-overlap labels are near-independent of DIF flags (enrichment $\approx$ 0.96). Across six benchmarks, 17.9–31.3% of items exhibit temporal instability associated with item properties; LLM benchmarks need psychometric monitoring alongside score reporting.

Limitations

Methodological scope. MH-DIF detects only uniform DIF; logistic regression reveals an additional 24.8% non-uniform DIF, making 31% conservative. More fundamentally, DIF detects *that* items function differently but cannot determine *why*—the signal may reflect contamination, capability change, or training methodology shifts. A definitive causal decomposition remains out of reach without access to model training data. The semi-synthetic framework uses binary response flipping (incorrect $\rightarrow$ correct) to simulate differential contamination, representing a worst-case uniform exploitation pattern. Real-world contamination involves partial memorization, reasoning shortcuts, and format-specific advantages; detection power under such realistic patterns is untested. The moderate ΔTPR $= +0.508$ at $p = 0.5$ reflects sensitivity to severe differential behavior; the framework's primary methodological contribution—cross-subject matching achieving ΔFPR $= 0.000$—is a matching strategy property independent of injection realism.

Examinee non-independence. The MH χ^2 test assumes independent examinees, yet LLMs sharing base models, training data, or architectures violate this assumption. Empirical mitigations partially address this: permutation simulation bounds LD-attributable FPR inflation to $<$0.001 pp, and graduated de-duplication (K=20: C% $= 29.1\%$) shows family structure does not inflate results. French and Finch (2010) show that MH maintains nominal size under moderate clustering in educational settings, consistent with our empirical findings; however, the extreme relatedness of some LLM families (e.g., hundreds of Llama-2 fine-tunes) exceeds the clustering levels studied in that work. Cluster bootstrap resampling at the family level (29 families with $\geq$3 members, 1,882 models, B = 1,000) yields a 95% CI of $[26.7\%, 53.2\%]$, 3.1$\times$ wider than the model-level CI $[29.4\%, 38.1\%]$; the lower bound remains far above zero, confirming robustness under family-level dependence.

Matching boundary and sample size. Cross-subject matching assumes contamination affects a minority of domains; when contamination spans $>$10 subjects ($>$18% of MMLU), ΔFPR rises to 0.026–0.069. Model-level bootstrap may underestimate CI width due to within-family correlation; cluster bootstrap at the model-family level is a natural extension (the K=20 de-duplication analysis at C% $= 29.1\%$ provides a conservative robustness check). The GSM1K correlation analysis ($N = 33$) has limited statistical power and is exploratory.

Generalizability. All six benchmarks use multiple-choice or binary scoring; generalizability to open-ended generation (e.g., AlpacaEval), coding (e.g., HumanEval), or partial-credit benchmarks is untested and would require polytomous DIF methods. Temporal cohorts are defined by calendar year, a coarse grouping; finer-grained definitions (e.g., by training data cutoff or alignment method) might reveal more targeted patterns.

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A Method Details

A.1 Diagnostics

Three diagnostic procedures complement the primary DIF analysis.

Local independence (Yen's Q3). The MH procedure assumes that item responses are conditionally independent given θ. Yen's Q3 statistic measures residual correlations between item pairs after conditioning on the latent trait. We compute Q3 for all within-subject and cross-subject item pairs, with $|Q3| > 0.20$ as the threshold for local dependence. Elevated within-subject dependence would indicate that items within the same subject share variance beyond what θ captures, which could inflate within-subject DIF rates.

Domain-specificity. We test whether temporal DIF concentrates in domains where models improved most. At the ecological level, we compute Spearman ρ between subject-level C% and subject-level accuracy improvement from 2023 to 2024. At the item level, logistic regression predicts item-level DIF C classification from domain-level accuracy change, with and without item property controls (difficulty, discrimination). This decomposition quantifies how much of the observed instability is attributable to aggregate domain-level gains versus item-specific behavioral change.

Instability decomposition. When domain accuracy improvement is included alongside item-level controls, we compute the counterfactual C% that would remain after removing domain-level effects. The gap between observed and counterfactual C% measures the contribution of domain-level improvement to the total instability.

A.2 Statistical Considerations

Multiple testing. The ETS C classification imposes a dual threshold (effect size $|\Delta_{\mathrm{MH}}| \geq 1.5$ and significance $p < 0.05$), which provides implicit control over false positives. As a robustness check, we apply Benjamini-Hochberg FDR correction at $q = 0.05$. The random-split sanity check provides an empirical estimate of the false positive rate under the null hypothesis.

Threshold sensitivity. We report results across a range of DIF thresholds ($|\Delta_{\mathrm{MH}}| \in \{1.0, 1.5, 2.0, 2.5\}$) to verify that qualitative conclusions do not depend on the specific ETS C cutoff. The standard threshold of $|\Delta_{\mathrm{MH}}| \geq 1.5$ corresponds to the ETS C definition used throughout the paper.

IRT model fit. For transparency, we report that the MMLU response matrix shows poor fit to a unidimensional 2PL model (expected for a 57-subject multidimensional benchmark). This does not affect MH-DIF validity, which relies only on total score stratification rather than parametric IRT assumptions.

Architecture heterogeneity. The METABENCH MMLU population consists of 99.98% decoder-only models (5,226 of 5,227). This near-complete architectural homogeneity eliminates architecture as a confound for DIF analysis. We verify this by excluding the single encoder-decoder model and confirming identical results.

Power sensitivity to item discrimination. The Monte Carlo power analysis in Section 4.1 uses average 2PL parameters ($\bar{a} = 1.2$, $\bar{b} = -0.84$). Table 4 reports TPR as a function of discrimination a (other parameters at their means, $N = 80$ per cohort, $|\Delta_b| = 1.4$, 1,000 replications). TPR increases monotonically from 0.40 ($a = 0.5$) to 0.95 ($a = 2.0$); FPR remains below 0.04 throughout. The MMLU item pool shows substantial heterogeneity: a ranges from 0.006 to 2.657 (SD $= 0.54$, IQR $= [0.75, 1.64]$). The aggregate TPR of 0.68 from Phase 2 reflects this full distribution, while TPR at mean $a \approx 1.2$ is 0.87—the gap is consistent with Jensen's inequality applied to the concave TPR(a) function. Although 2PL fit is poor for MMLU's multidimensional structure, the power analysis serves as one leg of a triangulation: parametric estimates (TPR $= 0.68$–0.80) are validated by empirical semi-synthetic injection ($\Delta\mathrm{TPR} = +0.508$ at $p = 0.5$; Section 4.5), confirming adequate detection power regardless of parametric assumptions.

Table 4: Power sensitivity to item discrimination. TPR = true positive rate for ETS C detection ($|\Delta_{\mathrm{MH}}| \geq 1.5$) at $N = 80$ per cohort, $|\Delta_b| = 1.4$, 1,000 replications.

a	TPR	95% CI	FPR
0.5	0.398	[0.20, 0.60]	0.033
0.8	0.683	[0.45, 0.90]	0.036
1.0	0.808	[0.60, 0.95]	0.038
1.2 ($\approx \bar{a}$)	0.874	[0.70, 1.00]	0.038
1.5	0.930	[0.80, 1.00]	0.038
2.0	0.949	[0.80, 1.00]	0.037

Logistic regression specification. The Simpson's paradox analysis (Section 4.2) uses binary logistic regression with DIF-C classification as the dependent variable, domain-level accuracy change (2024 minus 2023 mean accuracy) as the primary predictor, and item difficulty (proportion correct) and IRT discrimination (point-biserial correlation) as controls. The interaction between domain accuracy change and item difficulty is included. Non-uniform DIF detection (Section 4.4) uses logistic regression with a group $\times$ ability interaction term; a significant interaction ($p < 0.05$) indicates non-uniform DIF.

B Structural Analysis Details

Three progressively refined approaches fail to reduce the temporal false positive rate below 31%, revealing a structural property of the data rather than a methodological limitation. Matched-ability quintile analysis restricts comparisons to models of similar total score and yields enrichment ratios of 0.94–1.04, with temporal FPR unchanged at approximately 0.31. IRT θ-conditioning with 20 ability strata achieves $r(\theta, \text{total score}) = 0.9986$ and FPR $= 0.312$, confirming that the external total score already captures nearly all ability variance relevant to DIF computation. Within-subject DIF analysis on 5 representative subjects, intended to isolate subject-specific effects, finds mean C% $= 38.1\%$, exceeding the 25% threshold that would indicate viable bifactor structure. Table 2 summarizes these results.

The temporal FPR reflects genuine item-level instability that persists regardless of how ability is measured or conditioned. Yen's Q3 diagnostic reveals mild local dependence within subjects (11.3% of within-subject item pairs exceed $|Q3| > 0.20$, vs. 2.2% cross-subject; $5\times$ ratio, $p = 2.3 \times 10^{-9}$), which contributes to elevated within-subject DIF rates but is not the primary driver: the cross-subject matching protocol addresses this bias, and temporal C% persists under external matching (full Q3 distributions available in the supplementary material).

Table 5: Multi-subject injection robustness. k = number of simultaneously contaminated subjects (out of 57); values are mean $\pm$ SD across 5 random subject draws at exploitation rate $p = 0.5$.

k	ΔFPR	ΔTPR
1	$0.000_{\pm.000}$	$0.371_{\pm.048}$
3	$0.003_{\pm.002}$	$0.385_{\pm.031}$
5	$0.007_{\pm.003}$	$0.392_{\pm.027}$
10	$0.047_{\pm.022}$	$0.401_{\pm.024}$
20	$0.069_{\pm.018}$	$0.410_{\pm.020}$

C Cross-Benchmark Comparison

D Additional Figures

Table 6: Cross-benchmark comparison of DIF analysis on MMLU and GSM8K. Both benchmarks exhibit temporal measurement instability with validated semi-synthetic detection power. Standard and external matching use different exploitation rates (p) because each benchmark's optimal operating point differs; external matching results are compared at $p = 0.5$ for both. Subscript $\pm$ values denote standard deviations across 5 random injection seeds.

Metric	MMLU	GSM8K
Items	12,508	1,319
Models	5,227	6,702
Temporal C%	31.3%	18.9%
Sanity C%	0.18%	0.3%
ΔTPR (p=0.7, standard)	–	$+0.818_{\pm.006}$
ΔTPR (p=0.5, external)	$+0.508_{\pm.027}$	$+0.514_{\pm.028}$

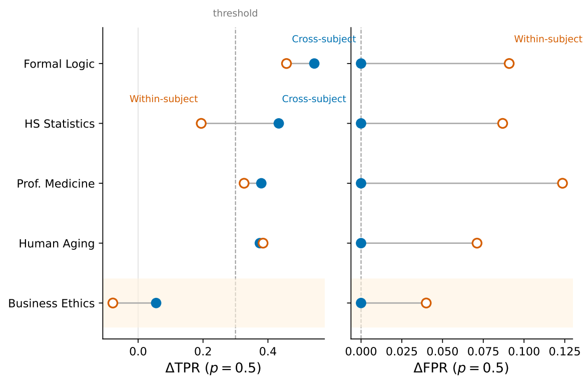

Figure 3: Cross-subject vs. within-subject matching for semi-synthetic DIF detection. Cross-subject matching eliminates FPR inflation ($\Delta\text{FPR} = 0.000$) while preserving detection power ($\Delta\text{TPR} > 0.3$ for 4/5 subjects at $p = 0.5$). Within-subject matching inflates FPR by $+0.186$ on average.

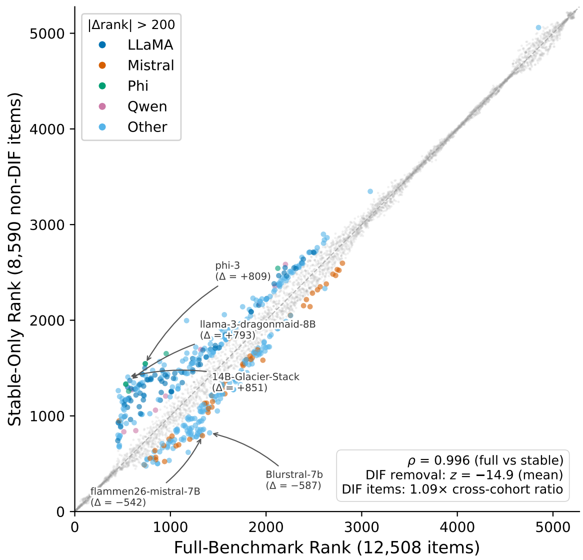

Figure 4: Downstream impact of DIF-based item filtering on model rankings. Despite $\rho \approx 0.997$ overall, individual model families show systematic rank shifts when evaluated on stable items only.

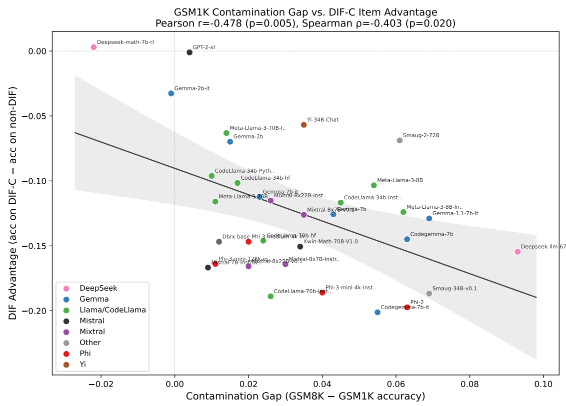

Figure 5: GSM1K contamination gap vs. DIF ranking advantage ($N = 33$). The negative correlation ($r = -0.478$, $p = 0.005$) indicates that models with larger contamination gaps benefit less from DIF-flagged items.